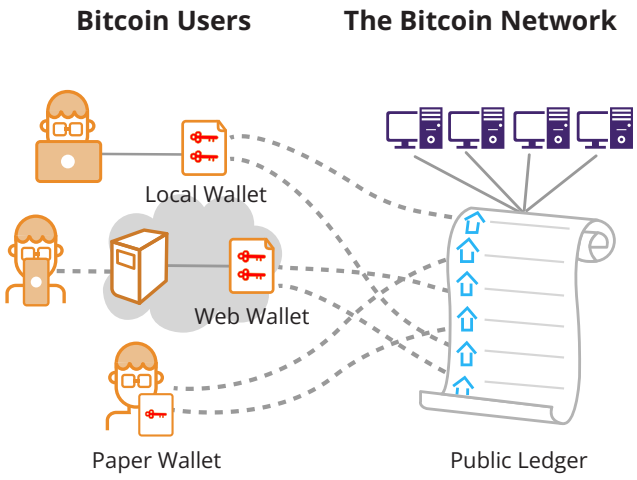
How Does Cryptography Work in Bitcoin Transactions?

In traditional private-key cryptography, John and Arel would have to first share a secret key, like a password. John would then use this key to scramble his message before sending it to Arel. Arel, who also knows the secret key, would then use the same key to unscramble the message and read it.

However, Ronny could also intercept the message and use the same key to unscramble it and read the message.

With **public key cryptography**, which is the type of cryptography used in bitcoin transactions, John and Arel each have two keys: a **public key** and a **private key**. John can use Arel's **public key** to scramble his own message before sending it (to Arel). Only Arel's **private key** can unscramble the message. Ronny, who does not have Arel's **private key**, would not be able to read the message even if he intercepts it.

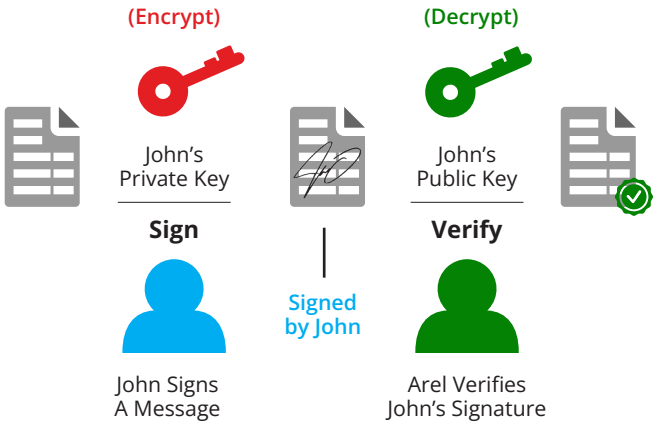


In addition to encrypting messages, **public key** cryptography can also be used for digital signatures. A digital signature is a way to prove the authenticity of a message, similar to a written signature on a physical document. In order to create a digital signature, John would use **his private key** to **encrypt his signature**. Arel then uses John's **public key** to decrypt it and verify that it was indeed sent by John.

Public Key Cryptography (For every transaction between two users):

Each user has two keys; a **private key**, which is kept secret, and a **public key** that can be shared with others.

The **private key** serves as a form of identification and proof of ownership, confirming: **"This address belongs to me and I have control over it"**

Digital signatures are created to identify unique transactions.

Digital Signature